Feasability Study

1. Introduction

Switzerland offers a wide range of trails for non-motorized travel targeted for leisure activities. Information on the officially marked routes is compiled on the SchweizMobil Foundation's platform. To gain an overview of which Swiss cantons are best suited as travel destinations for different target groups, this project examines how the cantons differ in terms of their trail offer by answering three research questions. Those questions address different target groups: travellers looking for hiking opportunities outside the summer season (RQ1), hikers with different levels of experience (RQ2), and visitors who want to combine various outdoor activities (RQ3). Findings might be relevant for local tourism offices in cantons which are leading in one of the research areas.

2. Research Questions

RQ 1: Which Swiss cantons provide the highest density of outdoor routes under 1200m maximum altitude that remain fully accessible during the spring and autumn shoulder season?

Approach: The route density is measured as the number of routes starting or ending in the canton compared to the area of the respective canton. The threshold of 1200m is chosen because according to MeteoSwiss (1991-2029)1, 1200 m serves as an empirical boundary for the freezing level during spring and autumn. Classifications by SAC2 define 1200 m as the upper limit of the montane zone.

RQ2: Does the percentage of hiking trails for each difficulty level relate to the mean elevation of a canton?

Approach: All hiking trails are classified by a grading scheme that considers how technically challenging they are. There are three difficulty levels: easy (hiking trails), medium (mountain hiking trail), and difficult (alpine hiking trails). To answer this question, the percentage of each difficulty type per canton is calculated and then compared to the mean altitude of the canton.

[1] MeteoSwiss: "Jahresverlauf der Nullgradgrenze" https://www.meteoschweiz.admin.ch/klima/klima-der-schweiz/jahresverlauf-temperatur-sonnenschein-und-niederschlag/jahresverlauf-der-nullgradgrenze.html

[2] *Schweizer Alpen-Club SAC:* EnviroTools Höhenstufen https://www.sac-cas.ch/fileadmin/Umwelt/Bergsport_und_Umwelt/EnviroTools/EnviroTools_H%C3%B6henstufen.pdf

RQ 3: Which Swiss cantons offer the best opportunities for combining different outdoor activities, measured by the density of routes within each activity type (hiking, cycling, mountain biking, skating and canoeing)?

Approach: For answering this question, the chapters "Wanderland", "Veloland", "Mountainbikeland", "Skatingland" and "Kanuland" on schweizmobil.ch are scraped for routes, respectively sections of large routes. For each canton, the route density (number of routes in relation to the total surface of the canton) per activity type is calculated. As the number of routes differs strongly between activity types (e.g. hiking vs. canoeing), the densities cannot be compared directly. Therefore, each density is converted into a percentile rank across all 26 cantons, where 1 indicates the highest density for that activity. The mean of the five percentile ranks results in a combined index.

3. Data Sources

- **Primary Source (Dynamic Web Scraping with Selenium):** schweizmobil.ch **Extracted Variables:** Route ID, title, activity modality, total distance, elevation gain/loss, maximum altitude, duration, starting point, ending point, difficulty level. The robots.txt of schweizmobil.ch allows all user agents to access all pages.
- **Geographical Information on the Cantons** ("Geographische Kennzahlen"): includes average elevation and area of the canton
- **Amtliches Ortschaftenverzeichnis, AMTOVZ** (downloaded from swisstopo.ch): The columns needed for matching start and endpoint of the routes to a canton are "Ortschaftsname" and "Kantonskürzel".

4. Risks

Risk	Impact	Migitation Strategy
Unknown if maximum altitude of the route can be scraped	RQ 1	Estimate the maximum altitude with a lower bound (higher start and end altitude) and an upper bound (start altitude + total elevation gain).
Cantons crossed by routes or route sections are not considered.	RQ 3	Accepted as a limitation: routes are assigned to the cantons of their start and end points; this simplification is discussed in the final documentation.
Matching the communes to the cantons	all RQ	Matching based on "Amtliches Ortschaftenverzeichnis"; handling manually if necessary.
Small cantons like BS or GE might show extreme density values	RQ 1 & RQ 3	Discuss spererately or mark as outliers